Application Note: AN1200.92

LR1121 Modem-E Evaluation Guide

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1 Introduction

This application note explains how to evaluate the LoRa Basics™ Modem-E firmware for the LR1121 chip using the evaluation tools provided by Semtech. It describes:

- LoRa Basics Modem-E firmware for the LR1121
- Main differences with the LR1110 Modem-E firmware
- Where to find and how to use the various HW and SW evaluation tools:
  - LR1121 Evaluation Kit
  - Software examples
  - Command interpreter for PC

2 LR1121 LoRa Basics™ Modem-E

2.1 Overview

The LoRa Basics™ Modem-E is a software that supports the LoRaWAN® Standard modem stack. It is embedded inside the LR1121, thus removing the need to integrate a LoRaWAN protocol stack running on the host MCU.

A small memory footprint host MCU can be chosen to call high level modem APIs and run the application software, thus LoRa Basics™ Modem-E allows LoRaWAN applications to be developed with the fastest time to market and a low overall BOM price.

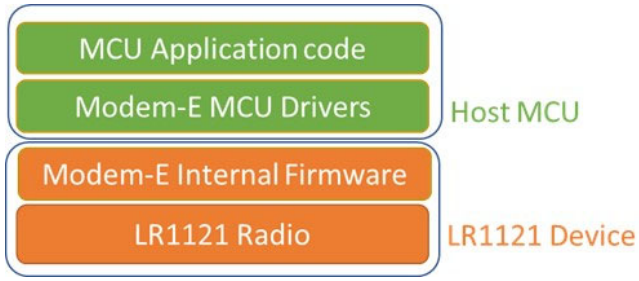

Figure 1: LoRa Basics™ Modem-E Components

LoRa Basics Modem-E firmware also offers the possibility to run all LR1121 transceiver commands with the same modem firmware (without the need to reprogram the LR1121 internal firmware), simply by putting the modem in suspend mode.

Using LoRa Basics Modem-E firmware for a product development allows also to leverage Semtech's LoRaWAN certification, and benefit from certification by similarity to reduce the LoRaWAN certification cost and effort.

2.2 Features

As the LoRa Basics Modem LoRaWAN stack is integrated inside the LR1121 chip, the LR1121 LoRa Basics™ Modem-E is aligned with the LoRa Basics Modem master branch version and supports the following features:

- LoRaWAN® Class A, B and C Standards version 1.0.4
- Regional parameters:
  - EU868
  - US915
  - AS923_GRP1
  - AS923_GRP2
  - AS923_GRP3
  - AS923_GRP4
  - AU915
  - CN470
  - CN470_RP_1_0
  - IN865
  - KR920
  - RU864
  - WW2G4
- Multicast class B and C. Up to 2 remote Multicast sessions.
- Transceiver commands are supported
- Embedded certification mode
- Relay end-device

- Listen Before Talk (LBT)
- Carrier Sense Multiple Access (CSMA)
- Firmware Update Over the Air (FUOTA) packages, experimental release
    - one fragmentation session at a time is supported
    - Updates are limited to 30KB

The LoRa Basics™ Modem-E firmware also allows to access LR1121 Transceiver commands for direct access to the LR1121 radio.

2.3 LR1110 versus LR1121

The LR1110 LoRa Basics Modem-E has been available since 2020 but will not be developed further and is not recommended for designs that utilize the GNSS scanning features of the LoRa Edge™ platform. Customers are advised to refer to the LoRa Basics™ Modem repository for applications requiring LoRa Edge geolocation: *https://github.com/Lora-net/SWL2001*.

The LR1121 LoRa Basics Modem-E firmware is recommended for future use for the following reasons:

- More features (see next section)
- Aligned with LoRa Basics Modem
- Offers a modem bypass mode to run LR1121 in transceiver without reflashing the internal firmware

LoRa Basics Modem-E firmware for LR1121 does not support geolocation.

3 LR1121 Evaluation Kit (EVK)

The LR1121 Evaluation Kit is based on an LR1121 evaluation shield PCB, plugged on an STM32L476 RG Nucleo kit. It provides all the necessary hardware to use and evaluate the LoRa Basics Modem-E.

3.1 Host MCU Drivers

Semtech provides LoRa Basics Modem-E drivers for the LR1121 EVK host MCU (STM32L476 RG) to ease application software development.

3.2 Embedded Firmware

The LoRa Basics Modem-E firmware runs inside the LR1121 internal flash.

The LR1121 device is flashed during production with the default Transceiver firmware image.

The LoRa Basics Modem-E firmware has then to be loaded inside the LR1121, for example using Semtech updater tool running on the LR1121 Evaluation kit, or in the customers production line.

The LoRa Basics Modem-E firmware and the updater tool are available on Github/LoRa-net (refer to section 4.2).

AN1200.57 provides guidance on how to program the LR1121 internal program memory.

3.3 EVK Hardware

The LoRa Basics Modem-E uses the LR1121 Evaluation Kit (EVK) hardware which already has the production firmware image embedded by default. It is necessary to flash the LoRa Basics Modem-E firmware using the LoRa Basics Modem-E updater tool.

The LR1121 Evaluation Kit is composed of:

- 1x LR1121 radio shield, PCB E655v01a
- 1x STM32L476RG Nucleo board
- 1x 2.4 GHz antenna with SMA connection
- 1x region specific sub-GHz antenna with SMA connection (915 MHz, 868 MHz or 490 MHz frequency bands)
- 1x read-me note

Figure 2: LR1121 Evaluation Kit Content

The following LR1121 Evaluation kits references are available for the different regions:

Reference	Description
LR1121DVK1TBKS	LoRa 868MHz ISM Band for Europe and 2.4GHz ISM Band
LR1121DVK1TCKS	LoRa 915MHz ISM Band for North America and 2.4GHz ISM Band
LR1121DVK1TGKS	LoRa 490MHz ISM Band for China and 2.4GHz ISM Band

Table 1: LR1121 Evaluation Kit References

3.4 LR1121 Shield Design Files

LR1121 shield design files can be found in the "Reference Design" Section of the LR1121 product page on Semtech website:
https://www.semtech.com/products/wireless-rf/lora-connect/lr1121#documentation
They contain the electrical schematic, PCB layout and Bill of Material, as well as the Shield Altium design files.

4 LoRa Basics Modem-E Usage & Evaluation

4.1 LR1121 & Modem-E Documentation and Software Overview

The figure below shows the various hardware, software, and tools that allow a customer to use and evaluate LoRa Basics Modem-E.

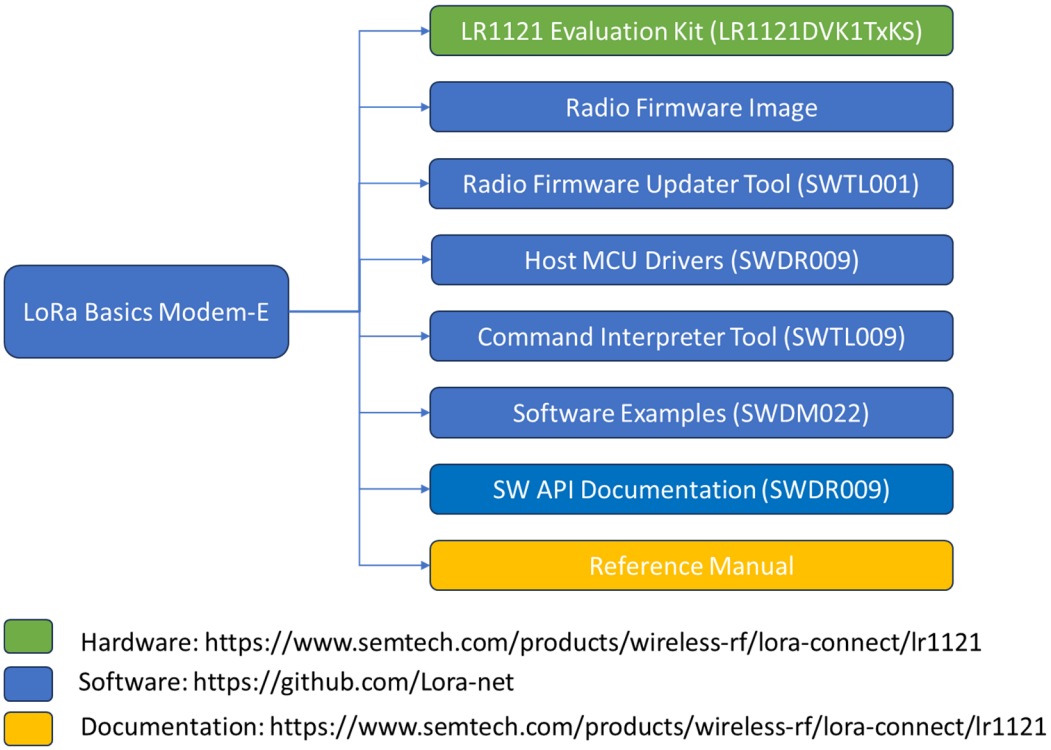

Figure 3: Documentation & Software

4.2 LoRa Basics Modem-E Firmware

The LoRa Basics Modem-E internal firmware is available in binary format at the following location:

https://github.com/Lora-net/radio_firmware_images/tree/master/lr1121/modem

The LR11xx updater tool allows to easily update the LR1121 radio firmware image on the Evaluation Kit, by a simple drag and drop of the .bin file in the Nucleo board enumerated as a mass storage device. It is available at the following location:

https://github.com/Lora-net/SWTL001

4.3 Reference Manual

The LoRa Basics Modem-E reference manual provides a detailed description of the LoRa Basics Modem-E firmware.

It is available on the LR1121 product page on Semtech.com website: https://www.semtech.com/products/wireless-rf/lora-connect/lr1121

4.4 Software API and Modem-E Driver

The LoRa Basics Modem-E API are described through the documentation of its driver, embedded as Doxygen documentation. The Modem-E driver can be found under https://github.com/Lora-net/lr1121_modemE_driver

4.5 Example Code

Semtech provides code that shows how to implement a LoRa Basics Modem-E application at the host MCU level, developed on the LoRa Basics Modem-E Evaluation kit.

The following example code is available on https://github.com/Lora-net/SWDM022

- Request TX
- Class B
- Multicast Class B
- Multicast Class C
- LCTT Certification
- FUOTA

4.6 Modem-E Command Interpreter Tool

The LoRa Basics Modem-E Command Interpreter Tool sends high level modem commands to the LoRa Basics Modem-E directly from a PC serial interface. It allows to quickly evaluate the LoRa Basics Modem-E without the need to develop any host MCU software.

The Modem-E Command Interpreter Tool allows to send high level Modem commands directly from a PC serial interface.

It is available under https://github.com/Lora-net/SWTL009

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Set up instructions:

1. Plug the LR1121 shield to the STM32L476 Nucleo board and connect the Nucleo board to the PC via a USB cable. The Nucleo board is then enumerated as a mass storage disk, and visible in the file explorer.
2. Ensure that the antennas are connected to allow the best radio performance.
3. Ensure that the LR1121 is programmed with the latest LoRa Basics Modem-E firmware. Instructions to upgrade the LR1121 to Modem-E are provided in Section **Error! Reference source not found.**.
4. Flash the Modem-E-interpreter-binary STM32L476.bin to the stm32l476 Nucleo board (drag and drop the .bin file on the as a mass storage disk in the file explorer)
5. Use a terminal serial tool like Mobaxterm or Putty to communicate with the right serial port, and set the datarate to 115200bps

Tips:

- Use 'help' for the list of available commands
- Arguments are separated by whitespace. For example: set_dev_eui FE FF FF FF FD FF 3B 00
- Numbers are usually treated as hex value unless noted in help info. For example: "0x11 0x3B" = "11 3B" = "0x11 3B"
- Commands are case-insensitive.
- Use “find” to filter commands, such as “find join” or “find lbt set”
- Use TAB key for completing commands, as well as basic readline (https://loicpefferkorn.net/pdf/readline_shortcuts_a4.pdf) shortcuts.
- Input commands are available as history, so use PgUp/PgDown or “Ctrl-P/Ctrl-N” to recall them.

A detail of the functions supported in the PC interpreter tool is shown below, and accessible with the “list_cmds” or “‘help” commands:

```
COM24 - Tera Term VT
File  Edit  Setup  Control  Window  Help

Use 'list_cmds' to list all commands.
Or 'find str' to search all commands whose names contain 'str'.

>> list_cmds
Available commands are:
class_b_get_ping_slot_periodicity
class_b_set_ping_slot_periodicity
csma_get_parameters
csma_get_state
csma_set_parameters
csma_set_state
derive_keys
find
fuota_get_file
fuota_get_file_size
get_adr_profile
get_class
get_dev_eui
get_downlink_data
get_downlink_data_size
get_downlink_metadata
get_duty_cycle_status
get_enabled_datarate
get_join_eui
get_lorawan_time
get_lorawan_version
get_nb_trans
get_network_type
get_next_uplink_max_payload
get_region
get_relay_config
help
join_network
lbt_get_params
lbt_get_state
lbt_set_params
lbt_set_state
leave_network
list_cmds
lorawan_get_lost_connect_count
lorawan_mac_request
modem_factory_reset
modem_get_charge
modem_get_crashlog
modem_get_event
modem_get_status
modem_get_version
multicast_class_b_get_session_status
multicast_class_b_start_session
multicast_class_b_stop_all_sessions
multicast_class_b_stop_session
multicast_class_c_get_session_status
multicast_class_c_start_session
multicast_class_c_stop_all_sessions
multicast_class_c_stop_session
multicast_get_group_config
multicast_set_group_config
request_emergency_uplink
request_empty_uplink
request_uplink
set_adr_profile
set_app_key_multicast
set_class
set_dev_eui
set_join_datarate_distribution
set_join_eui
set_nb_trans
set_network_type
set_nwk_key
set_region
set_relay_config
system_reboot
test_exit
test_nop
test_read_rssi
test_rssi
test_start
test_tx_cw

>>
```

Below are a few example instructions to be issued to join the network, upload some data, run test_tx_cw, run test_rssi and test_read_rssi

4.6.1 LoRaWAN join

```
>> set_dev_eui  FE FF FF FF FD FF 3B 00
>> set_join_eui  0x0, 0x16, 0xC0, 0x01, 0xFF, 0xFE, 0x00, 0x01
>> set_nwk_key  0 0 0 0 0 0 0 0 0 0 0 0 0 0 3B 0
>> join_network
>>
----> Event happened (0 such event missed): JOINED
```

4.6.2 LoRaWAN Uplink

```
>>request_uplink -h
>>
>> request_uplink 0x20 unconfirm 40 50
>> leave_network
----> Event happened (0 such event missed): DOWNDATA
data size: 5
data: 0x55 0x66 0x77 0x88 0x99
meta data: 0x00 0xDA 0x18 0x01 0x07 0x00 0x33 0xC1 0x34 0xE0 0x00
meta data details:
  - stack id: 0
  - rssi: -102
  - snr: 6.00
  - window: Received in 1st downlink slot
  - fport: 7
  - frame pending bit: 0
  - frequency (Hz): 868300000
  - data rate: 0
```

4.6.3 Test_Tx_CW

```
>> test_start
>> test_tx_cw 868300000 10
>> test_exit
>>
----> Event happened (0 such event missed): TEST_MODE
STATUS: Test mode ended action
```

4.6.4 RSSI

```
>> test_rssi 868900000 10 125000
>> test_read_rssi
rssi: -120
>> test_rssi 867200000 10 125000
>>
>> test_read_rssi
rssi: -119
>> test_rssi 866200000 10 125000
>> test_read_rssi
rssi: -118
>> test_exit
```

5 Revision History

Version	ECO	Date	Changes and/or Modifications
1.0	ECO-072867	September 2024	Initial Version

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